

Parachute: Evaluating Interactive Human-LM Co-writing Systems

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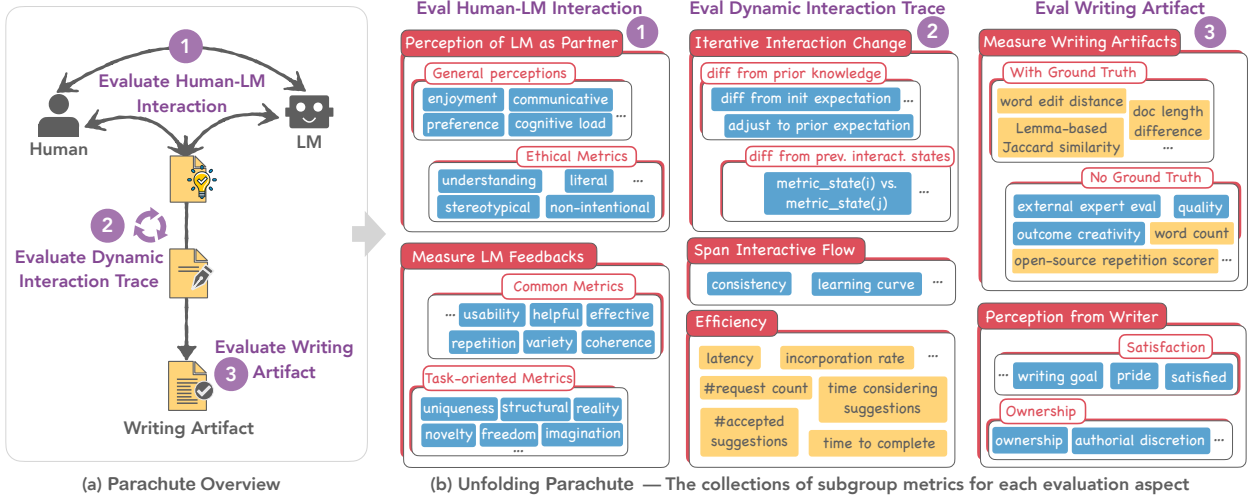


Figure 1: Parachute: human-centered integrative evaluations on interactive co-writing systems. (a) Parachute overview. (b) The subgroup metrics for each evaluation aspect. (yellow and blue indicates objective and subjective metrics, respectively.)

ABSTRACT

A surge of advances in language models (LMs) has led to significant interest in using LMs to build co-writing systems, in which humans and LMs interactively contribute to a shared writing artifact. However, there is a lack of studies assessing co-writing systems in interactive settings. We propose a human-centered evaluation framework, Parachute, for interactive co-writing systems. Parachute showcases an integrative view of interaction evaluation, where each evaluation aspect consists of categorized practical metrics. Furthermore, we present Parachute with a use case to demonstrate how to evaluate and compare co-writing systems using Parachute.

1 INTRODUCTION

Language models (LMs) have advanced significantly, showcasing previously unheard-of capabilities in solving a wide spectrum of generation and language understanding tasks [3, 8, 9]. This has spurred great academic and public interest in using LMs to build writing assistants, in which humans collaborate with LMs to paraphrase sentences (e.g., QuillBot), autocomplete sentences [2], write stories [1], etc. Despite the interactive co-writing process, the co-writing systems are at present primarily tested in non-interactive settings [4, 12]. Specifically, current studies commonly conduct one-time evaluations on writing article quality [7], or prior- and post-human assessment on perceiving LMs [11, 12], etc. As a result, these evaluations fall short of capturing the dynamic shift in human-LM interactions and even fair assessments. For instance, suppose a scientific paper writing task involves two users, a freshman and a university professor, to participate in user study evaluations. With the help of LMs, the freshman makes much improvement in writing

papers by benefitting from LMs’ suggestions (e.g., understanding the paper structure, grammar correction, etc.) While the professor is skilled at writing in general, the system is considerably less helpful for him. However, when making the one-time evaluation of the final paper’s quality, the professor’s performance will likely outperform the freshman’s significantly. This result can not genuinely reflect the co-writing system’s influence on users. Therefore, we should instead assess users’ dynamic interaction improvements to indicate system capability, such as measuring relative quality change between two iterated articles from the same user.

Nevertheless, as far as we know, there is at present a lack of studies in examining *how to assess the interaction shift along the iterative co-writing process*, and furthermore, *how to depict an integrative view for evaluating interactive co-writing systems?* To close this gap, we propose Parachute: a human-centered evaluation framework for human-LM interactive co-writing systems. We identify the key components and interaction aspects for evaluating the co-writing systems. We further collect the comprehensive types of evaluation metrics under each aspect, including the novel measurements designed for dynamic interaction assessments, and the other two conventional yet important aspects (i.e., human-LM interaction and writing artifact evaluation). Additionally, we present a use case study to exemplify how to leverage Parachute as a thinking tool to comprehensively evaluate and compare the co-writing systems.

2 PARACHUTE: A THINKING TOOL TO GUIDE INTEGRATIVE INTERACTION EVALUATION

A human-LM co-writing system involves both human and LM collaborating on a shared writing artifact (e.g., essay, story, paper,

etc.) as partners [10]. Grounded on the Co-Creative Framework for Interaction Design (COFI) model [10], we identify three key **components** in co-writing systems, including **human**, **LM**, and **writing artifact**. They act as the joint driving force of the interactive co-writing process. As humans are the essential decision-makers to interact with LMs and write artifacts, in this work, we propose a human-centered evaluation framework¹, Parachute, to comprehensively evaluate the human-LM interactive co-writing systems.

As described in Figure 1, Parachute further recognizes three crucial **aspects** for evaluating the components' interactions, including ① **evaluate human-LM interaction**, ② **evaluate dynamic interaction trace**, and ③ **evaluate writing artifact**. Compared with existing evaluation approaches, which might capture insufficient evaluation aspects to reflect the system capability holistically [4, 7, 12] or focus much on one-time metrics that neglect dynamic interaction changes [5, 11], Parachute seeks to contribute in two folds. First, Parachute presents a more integrative view of interaction evaluation supported by practical metrics. Besides, Parachute explicitly extracts a set of metrics to assess the dynamic metric change along iterative interactions. Overall, we propose Parachute as a thinking tool, assisting researchers to practically evaluate and analyze interactive co-writing systems more comprehensively and fairly.

3 UNFOLD PARACHUTE: WHICH EVALUATION METRICS CAN WE USE IN PRACTICE?

To consolidate Parachute to be more useful in practice, we **unfold the three evaluation aspects into subgroups of practical evaluation metrics**. To achieve this, we use an inductive approach to collect the practical metrics adopted in state-of-the-art co-writing systems (e.g., *Wordcraft* [12], *Integrative Leaps* [11], *Beyond Text Generation* [4], *Dramatron* [7], etc.) Then we categorize them into subgroups and fit into Parachute framework. Figure 1(b) depicts the categorized evaluation metrics for each Parachute aspect. We next clearly define the aspects and metric subgroups in Parachute, and briefly explain the underlying motivation. Please see Table 1 in **Appendix A.1** for more elaborated metrics and details.²

① **Evaluating Human-LM interaction**. These metrics measure interactions between the co-writers (i.e., human & LM). Suppose humans perceive LM as a co-author in co-writing systems. Then their evaluations primarily derive from two dimensions: *i)* how does the human feel to collaborate with LM? (i.e., “**Perception of LM as Partner**”). This subgroup consists of both metrics of general perceptions (e.g., *enjoyment*, *preference*, etc.), and ethical metrics (e.g., *stereotypical*, etc.). Meanwhile, *ii)* how credible is the LM's feedback? (i.e., “**Measure LM Feedbacks**”), which involves common metrics for a variety tasks (e.g., *usability*, etc.), or task-oriented metrics, like *imagination* preferred by storytelling, or *structural* for scientific writing, etc.

② **Evaluating dynamic interaction trace**. These metrics focus on evaluating the dynamic change of interactions along iterative writing process. We identify three dimensions for evaluations. First,

when human iteratively updates the artifacts, “**Iterative Interaction Change**” subgroup aims to compare metrics between multiple iterations, such as measuring human understanding on LM *before* start writing and when almost *finish* the article). Also, we cover “**Span Interactive Flow**” subgroup here to assess metrics that need to observe spanning multiple artifact versions (e.g., *consistency*, *learning curve*). Besides, the responding time affects user experience in the interactive systems. We thus include “**Efficiency**” metrics (e.g., *latency*, *incorporation rate*) to assess the process.

③ **Evaluating writing artifact**. These metrics gauge the content of the final writing artifact that human and LM jointly accomplish. We broadly divide these metrics into “**Measure Writing Artifacts**”, where we can compare written artifacts with ground-truth articles (e.g., using *Jaccard similarity*) or recruit external experts for evaluation when without ground truth, and “**Perception from Writer**” dimensions, where writers provide subjective feedback on their outputs (e.g., *satisfaction* or *ownership*).

4 HOW TO USE PARACHUTE TO EVALUATE AND COMPARE CO-WRITING SYSTEMS?

We consider Parachute as a thinking tool for researchers to fairly evaluate and compare co-writing systems. Parachute can be useful for researchers to: 1) **identify key interactive evaluation aspects** among human, LM, and writing artifact interactions during the co-writing process, 2) **select appropriate metrics** to assess and compare the co-writing systems, 3) **comprehensively analyze and describe** the human-LM interactive performance of the co-writing systems. Next we present a concrete use case of re-evaluating the *Beyond Text Generation* (BTG) [4] system to showcase how to use Parachute for evaluation.

Suppose the researchers have built BTG system and need to evaluate its performance in the interactive setting. The researchers may now apply Parachute to conduct comprehensive evaluation on it and compare with existing baseline systems. For example, by checking the Parachute framework, *in the first step*, the researchers decide to validate the hypothesis: “**the BTG system can help humans write better articles** (i.e., *aspect3: evaluate writing artifact*) **by enabling better human-LM interactions** (i.e., *aspect1: evaluate human-LM interaction*) **in efficient ways** (i.e., *aspect2: evaluate dynamic interaction trace*). Next, they delve into each aspect to select the appropriate metrics to support this statement. For instance, they assess “writers' perceptions of LM” by choosing *enjoyable*, *preference* metrics, and “how writers think about LM's feedbacks” with *usability*, *effective*, *coherence* metrics, etc. Also, they assess the dynamic interaction efficiency by analyzing the objective logging data (e.g., *incorporation rate*, *latency*). Besides, they also evaluate the final writing article with a set of measures (e.g., *external expert review*, *quality*, *satisfaction*, *ownership*, etc.). Thirdly, after selecting these metrics, they design user study and approaches³ to accomplish these evaluations. Note that the researchers apply all measurements to both the proposed BTG system and the baselines, and ideally with the same group of users, so that they can compare the performance of co-writing systems in a fair manner.

¹“human-centered evaluation” means the metrics are designed to help humans achieve their various needs (e.g., better user experience, higher-quality writing artifact, etc.)

²Note that we do not aim to build enumerated lists of evaluation metrics. Instead, we focus on introducing the motivation and process of creating these evaluation aspects and subgroups, which can be generalized in broader use.

³In **Appendix A.2**, we present a list of evaluation methods, which are not comprehensive but commonly used in state-of-the-art co-writing systems' user studies [7, 11, 12], for collecting data and analyzing the interactive metrics.

5 CONCLUSION

This work present Parachute: a human-centered framework to evaluate human-LM interactive co-writing systems. It provides a thinking tool for researchers to design comprehensive interaction evaluations and analyses. We further feature a use case study introducing how to use Parachute step-by-step for fair evaluations.

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A APPENDIX

A.1 Parachute Metric Details

We list the practical evaluation metrics of Parachute with details, including Interaction Aspects, Subgroups, Metrics, Measure Questions, and References in Table 1.

A.2 Evaluation Methods in User Studies for Co-writing Systems

We summarize a list of evaluation methods that are commonly used in studying co-writing systems [7, 11, 12] for the future work reference. This summary aims to serve the purpose of providing inspirations and benchmarks for future co-writing system work to design and compare user study evaluations.

(a) **Coding users' think aloud transcripts.** Researchers can encourage the participants to articulate their thinking during interacting with the systems, such as *why they decided to use this prompt for querying LM but not others*, etc. After finishing the study, researchers can convert the video/radio into transcripts and code the transcripts for further data analysis. Some common qualitative data analysis approaches include thematic analysis, content coding, and topic modeling [6], etc.

(b) **Coding researchers' observation notes.** During the user studies, the researchers can also record their observations for later analysis. For instance, they can pre-design a set of topics (e.g., user's emotion change, etc.) that are important to answer their research questions, and pay close attention to these topics during the studies.

(c) **Coding (semi-)structured interview transcripts.** Prior studies also commonly leverage (semi-)structured interviews to elicit users' experience and feedback on using the systems. Researchers can design effective interview questions and invoke participants' answers, to better support the research arguments.

(d) **Questionnaires or Surveys.** Previous studies also frequently design surveys, which include questions such as N-five-point Likert ratings, single choice or multiple choice questions, etc. These surveys can provide more accurate measures on user's assessment.

(e) **Interaction data logs.** The data logs during interaction can provide more objective analysis on user behaviors. Typical interaction data logs for co-writing systems involve artifact submission count, prompt request frequency, latency, etc.

(f) **Assessment on the written artifacts.** These metrics aim to directly evaluate the quality of written artifacts, commonly using automatic metrics by comparing with ground truth (e.g., *similarity*), computing the artifact properties (e.g., *word count*, *document length*, etc.), or having external experts to assess the outputs.

Interaction Aspects	Subgroups	Metrics	Measure Questions	References
❶ Evaluating Human-LM Interaction	Perception of LM as Partner [General perceptions]	enjoyment effort preference	<i>I enjoyed writing the story.</i>	[7, 12]
			<i>I put lots of effort into getting AI suggestions.</i>	[11]
			<i>I prefer the suggestions from the AI agent.</i>	[7]
		communicative cognitive load collaborative ease	<i>I can communicate well with the AI agent.</i>	[11]
			<i>Interacting with the AI agent requires much cognitive load.</i>	[11]
			<i>I feel the AI agent collaborative to work together.</i>	[7, 11, 12]
	Perception of LM as Partner [Ethical Metrics]	understanding literal	<i>The AI agent is easy to learn and work with.</i>	[7, 12]
			<i>I can understand the AI agent.</i>	[11]
		stereotypical non-intentional	<i>The AI agent's suggestions are literal.</i>	[7]
			<i>The AI agent's suggestions are stereotypical.</i>	[7]
	Measure LM Feedbacks [Common Metrics]	coherent variety helpful effective repetition usability combinatorial	<i>The AI agent does not have intentions to generate suggestions.</i>	[11]
			<i>The AI generations are coherent with prompts.</i>	[7, 11]
			<i>The AI agent's suggestion are various.</i>	[7, 11]
			<i>I found the AI agent helpful.</i>	[7, 11, 12]
			<i>The AI agent is effective at suggesting ideas.</i>	[12]
			<i>The AI agent generates repetitive suggestions.</i>	[11]
❷ Evaluating Dynamic Interaction Trace	Iterative Interaction Change [diff from prior knowledge]	different from initial expectation adjust to prior expectation	<i>The AI agent is useful for my writing.</i>	[11]
			<i>I feel the AI agent combines a broad set of information.</i>	[11]
	Iterative Interaction Change [diff from prev. interact. states]	uniqueness reality novelty freedom structural imagination unexpected	<i>The AI's suggestions are unique.</i>	[7, 11, 12]
			<i>emphThe AI's suggestion aligns with common sense.</i>	[7, 11]
	Span Interactive Flow	learning curve consistency flow and ordering	<i>The AI agent often generates unexpected suggestions.</i>	[11]
			<i>I feel the AI agent can express freely.</i>	[11]
	Efficiency	latency incorporation rate request count time considering suggestions #accepted suggestions time to complete	<i>The AI suggestions are structural.</i>	[7]
			<i>I feel the AI agent has much imagination.</i>	[11]
			<i>The AI suggestions are often unexpected to me.</i>	[7, 11]
			<i>what's the difference before the initial expression.</i>	[11]
			<i>I adjusted my expectation to prior ones.</i>	[11]
			<i>how does the suggestion integration change dynamically.</i>	[11]
❸ Evaluating Writing Artifact	Measure Writing Artifacts [With Ground Truth]	word edit distance lemma-based Jaccard similarity document length difference	<i>I can learn to use this system quickly.</i>	[11]
			<i>The AI generate consistent suggestions along interaction.</i>	[7, 11]
			<i>the flow and ordering of co-writing are smooth.</i>	[7, 11]
	Measure Writing Artifacts [No Ground Truth]	outcome creativity quality external expert evaluation word count open-source repetition scorer	<i>The elapsed time from human request to AI response.</i>	[4]
			<i>The rate of incorporating AI suggestions.</i>	[4]
			<i>The count of human requests.</i>	[4]
	Perception from Writer [Satisfaction]	writing goal pride satisfied	<i>The average time for human to consider AI suggestions.</i>	[4]
			<i>The count of accepted AI suggestions.</i>	[4]
			<i>The elapsed time for human to complete the task.</i>	[4]
	Perception from Writer [Ownership]	ownership authorial discretion	<i>The word edit distance between prior- and post- articles.</i>	[7]
			<i>The similarity of ground truth and outcome article.</i>	[7]
			<i>The difference between prior- and post- articles.</i>	[7]

Table 1: The practical subgroup metrics for each evaluation aspect in Parachute. We demonstrate the metric details of Interaction Aspects, Subgroups, Metrics, Measure Questions, and the corresponding References. (yellow and blue indicates objective and subjective metrics, respectively.)